Luna Tian

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Education

University of Pennsylvania | MSE, Computer Graphics & Game Technology — Aug 2026 – May 2028 (expected)
- Coursework: Interactive Computer Graphics (CIS 5600), Computer Animation (CIS 5620), 3D Modeling

University of California, Berkeley | B.A., Computer Science (GPA 3.76/4.00) — Jun 2020 – Aug 2022
- Coursework: Algorithms (CS 170), Artificial Intelligence (CS 188), Operating Systems (CS 162), Database Systems (CS 186), Computer Security (CS 161), UI Design (CS 160), Machine Structures (CS 61C), Data Structures (CS 61B)

Experience

Deloitte | Software Engineering Intern — Jun 2026 – Aug 2026
- Built an end-to-end enrollment-rate forecasting pipeline on **∼140K** ClinicalTrials.gov records, engineering 70+ features and validating with a time-based split (train on past trials, test on future ones) to mirror real planning use.
- Benchmarked linear, Random Forest, XGBoost, and CatBoost; CatBoost cut average forecast error **∼23%** vs. Random Forest.
- Ran feature ablations and presented findings to engineers and non-technical stakeholders in Agile sprints.

Nokia | Software Developer Co-op — Jan 2026 – May 2026
- Built automated integration and regression test suites for auth UI, backend integration, and backup/restore.
- Configured **Jenkins CI/CD** pipelines with version-specific Helm values to deploy **Kubernetes** test environments.
- Authored end-to-end Selenium functional tests for login, custom banners, Terms & Conditions, and CAPTCHA flows, validating user-facing behavior across failed logins, refreshes, logout, and re-login.
- Built Cursor Agent skills that find test gaps and draft scenarios, cutting time-to-first-scenario from **hours to minutes**.

humanID (Foundation for a Human Internet) | Tech Team Lead; Software Engineer Intern — Jan 2024 – Dec 2025
- Led backend and infrastructure initiatives for a privacy-first authentication platform, partnering with product and cross-functional teams to align auth workflows with product requirements.
- Re-architected a Stripe-integrated billing service from cron-based async jobs to real-time sync, improving data freshness from hours to real time and **cutting peak-load query latency by more than half**.
- Built CI/CD (GitHub Actions, AWS, Docker) and configured Cloudflare SSL/TLS, WAF, DDoS protection, and rate limiting.
- Built REST-backed onboarding workflows for a Discord verification bot (Python, Django) serving **10K+ users**, with role-based access control and real-time event handling; boosted new registrations by **83%** and weekly logins by **53%**.

Projects

Catmull-Clark Subdivision Mesh Editor | *C++, OpenGL, Qt* — Demo
- Built a **half-edge mesh editor** with n-gon OBJ import and SYM pointer linking, plus tools for live vertex and face edits, edge splitting, face triangulation, and hotkey-driven topology inspection.
- Implemented Catmull-Clark subdivision with **near-linear scaling**: each 4× increase in faces costs 4.4–4.8× the time, taking a 5.8K-face cow mesh to **1.1M faces in 6.2 s** total.

Noise and Post-Process Shaders | *C++, OpenGL, GLSL, Qt* — Demo
- Built a real-time viewer with an off-screen framebuffer **post-process pipeline**: Blinn-Phong and matcap surface shaders, vertex deformation, Sobel and Gaussian filters, Worley-noise glass, and animated fog from **domain-warped fBM**.

Music-Driven Spiky Chrome Sphere | *Unity, C#, Shader Graph* — Demo
- Built a real-time audio-reactive Unity experience: a **C# FFT audio driver** (RMS energy, spectral flux, envelope smoothing) drives a procedural shader with triplanar noise and vertex displacement.

FK/IK Walking Character | *C++, Unity* — Demo
- Built a C++ animation toolkit as a native Unity plugin with BVH motion playback and **limb, CCD, and pseudo-inverse IK solvers**; the character walks to clicked targets while foot IK plants its feet on uneven terrain.

Saliency-Aware Pruning for 3D Gaussian Splatting | *Python, gsplat, SAM* — Report
- Built a semantic-aware 3DGS pruning pipeline (SAM segmentation, union-find clustering, multi-view visibility scoring), cutting checkpoint storage by **up to 91.6%** (avg. ∼70%) across 7 Mip-NeRF 360 scenes.

NumC | *C, SIMD, Python/C API* — GitHub
- Built a NumPy-style C matrix library via the Python/C API; SIMD-optimized code **ranked #1 in class** for speed.

Skills

Languages: C++, Java, C#, Python, C, JavaScript, SQL, GLSL
Backend & Infra: REST APIs, Django, Docker, Kubernetes, Helm, AWS, Cloudflare, Linux, Git
Testing: integration, regression & end-to-end testing, Selenium, CI/CD (Jenkins, GitHub Actions), Agile/Scrum
Games & Graphics: Unity, OpenGL, GLSL, Qt, real-time rendering